Detailed Appendix

Full derivations for Q4 (cap) and Q5 (RTM bargain)

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Companion document to `problem1_v2.pdf`. The main paper presents streamlined versions of §A.1 (primitives, FOCs, bliss points) and §A.2 (firm's constrained programme); the unabridged derivations, with full KKT machinery and case-by-case argument, are reproduced below. Sections §A.3–§A.5 and Appendix B (§B.1–§B.7) are reproduced verbatim from the main paper.

1 Formal derivation of the Q4 cap results

1.1 Primitives, FOCs, and the three bliss points

The worker and firm payoffs are

$$V_w(h) = (1-\tau)wh - \tfrac{\alpha}{2}h^2, \qquad V_f(h) = Ah - \tfrac{\beta}{2}h^2 - wh.$$

Worker FOC. $V_w'(h) = (1-\tau)w - \alpha h = 0 \implies h^* = (1-\tau)w/\alpha$ (the worker's preferred hours absent any constraint).

Firm FOC. $V_f'(h) = A - \beta h - w = 0 \implies h^{\text{firm}} = (A-w)/\beta$ (the firm's unconstrained labour demand at wage w).

Joint-surplus FOC. Adding the two payoffs the wage transfer cancels and

$$W(h) = V_w(h) + V_f(h) = Ah - \tau wh - \tfrac{\alpha+\beta}{2}h^2, \qquad W'(h) = A - \tau w - (\alpha+\beta)h = 0 \implies h^{\text{soc}} = \frac{A-\tau w}{\alpha+\beta}.$$

At the calibration $(A, w, \tau, \alpha, \beta) = (2, 1, 0.4, 2, 2)$: $h^* = 0.30$, $h^{\text{soc}} = 0.40$, $h^{\text{firm}} = 0.50$, so the firm-power ordering $h^* < h^{\text{soc}} < h^{\text{firm}}$ holds.

1.2 Firm's constrained programme under a cap

In the firm-power baseline the firm chooses h to maximise its own surplus subject to (i) the regulatory cap $h \le \bar{h}$ and (ii) the worker's individual-rationality (IR) constraint $V_w(h) \ge V_w^0$, with the wage chosen so that IR binds.

Why IR enters the programme. Without IR the firm-power problem is degenerate (the firm could set w arbitrarily low and extract unbounded rents); IR pins the wage at the lowest level that satisfies the worker's outside option V_w^0, closing the model.

The Lagrangian is

$$\mathcal{L}(h,\mu,\lambda) = V_f(h) - \mu(h-\bar{h}) + \lambda\big(V_w(h) - V_w^0\big), \qquad \mu,\lambda \ge 0.$$

The FOC and KKT slackness conditions read

$$V_f'(h) + \lambda V_w'(h) = \mu, \qquad \mu(h-\bar{h}) = 0, \qquad \lambda(V_w(h) - V_w^0) = 0.$$

The wage is set so that IR binds, which fixes λ; the inner choice of h then collapses to $\max_h V_f(h)$ s.t. $h \le \bar{h}$. Two cases:

- **Cap slack** ($\bar{h} \ge h^{\text{firm}}$): $\mu = 0$, $h^{\text{eq}} = h^{\text{firm}}$.
- **Cap binding** ($\bar{h} < h^{\text{firm}}$): $h^{\text{eq}} = \bar{h}$ and $\mu = V_f'(\bar{h}) = A - w - \beta\bar{h} > 0$ (the shadow price of the cap is the firm's marginal value of an extra hour at the constrained equilibrium).

Combined: $h^{\text{eq}}(\bar{h}) = \min\{\bar{h},\, h^{\text{firm}}\}$. All subsequent surpluses are evaluated at this h^{eq} relative to the no-cap baseline h^{firm}.

1.3 Quadratic identities for the three surpluses

Any concave quadratic $g(h)$ with bliss point $h^\dagger$ and curvature $-\kappa$ satisfies

$$g(h) - g(h^\dagger) = -\tfrac{\kappa}{2}(h - h^\dagger)^2. \tag{1}$$

Apply (1) to V_f (bliss h^{firm}, curvature β) and to W (bliss h^{soc}, curvature $\alpha + \beta$):

$$\Delta V_f(\bar h) \equiv V_f(\bar h) - V_f(h^{\text{firm}}) = -\tfrac{\beta}{2}\big(h^{\text{firm}} - \bar h\big)^2 \le 0, \tag{2}$$

$$\begin{aligned}\Delta W(\bar h) \equiv W(\bar h) - W(h^{\text{firm}}) &= \big[W(\bar h) - W(h^{\text{soc}})\big] - \big[W(h^{\text{firm}}) - W(h^{\text{soc}})\big] \\ &= -\tfrac{\alpha+\beta}{2}\Big[(h^{\text{soc}} - \bar h)^2 - (h^{\text{soc}} - h^{\text{firm}})^2\Big]. \end{aligned}\tag{3}$$

For ΔV_w the reference point h^{firm} is *not* the worker's bliss, so we expand directly:

$$\begin{aligned}\Delta V_w(\bar h) &= V_w(\bar h) - V_w(h^{\text{firm}}) \\ &= (1-\tau)w(\bar h - h^{\text{firm}}) - \tfrac{\alpha}{2}\big(\bar h^2 - (h^{\text{firm}})^2\big) \\ &= (\bar h - h^{\text{firm}})\Big[(1-\tau)w - \tfrac{\alpha}{2}(\bar h + h^{\text{firm}})\Big],\end{aligned}\tag{4}$$

using the factorisation $\bar h^2 - (h^{\text{firm}})^2 = (\bar h - h^{\text{firm}})(\bar h + h^{\text{firm}})$.

1.4 Welfare-improving band: derivation of the optimality condition

For $\bar h < h^{\text{firm}}$, (3) gives

$$\Delta W(\bar h) > 0 \iff (h^{\text{soc}} - \bar h)^2 < (h^{\text{soc}} - h^{\text{firm}})^2 \iff |h^{\text{soc}} - \bar h| < h^{\text{firm}} - h^{\text{soc}},$$

using $h^{\text{firm}} > h^{\text{soc}}$. Removing absolute values:

$$-(h^{\text{firm}} - h^{\text{soc}}) < h^{\text{soc}} - \bar h < h^{\text{firm}} - h^{\text{soc}} \iff 2h^{\text{soc}} - h^{\text{firm}} < \bar h < h^{\text{firm}}.$$

Maximising (3) in $\bar h$ gives the peak $\bar h = h^{\text{soc}}$ with gain

$$\Delta W(h^{\text{soc}}) = \tfrac{\alpha+\beta}{2}\big(h^{\text{firm}} - h^{\text{soc}}\big)^2 = \tfrac{4}{2}(0.10)^2 = 0.020.$$

Band collapse at $\alpha = \beta$. Substituting the bliss-point formulas,

$$2h^{\text{soc}} - h^{\text{firm}} = \frac{2(A - \tau w)}{\alpha+\beta} - \frac{A - w}{\beta} \xrightarrow{\alpha=\beta} \frac{A - \tau w}{\beta} - \frac{A-w}{\beta} = \frac{(1-\tau)w}{\beta} = h^*.$$

At $\alpha = \beta$ the lower band-edge equals the worker's bliss h^*, so the band simplifies to $\bar h \in (h^*, h^{\text{firm}}) = (0.30, 0.50)$.

1.5 Worker-surplus band and the two thresholds

The sign of ΔV_w comes from (4). For $\bar h < h^{\text{firm}}$, the leading factor $(\bar h - h^{\text{firm}}) < 0$, so $\Delta V_w(\bar h) > 0$ iff the bracket is also negative:

$$(1-\tau)w < \tfrac{\alpha}{2}(\bar h + h^{\text{firm}}) \iff \bar h + h^{\text{firm}} > \frac{2(1-\tau)w}{\alpha} = 2h^* \iff \bar h > 2h^* - h^{\text{firm}}.$$

Combined with $\bar h < h^{\text{firm}}$ (cap binding), the worker-surplus band is

$$\Delta V_w(\bar h) > 0 \iff \bar h \in \big(2h^* - h^{\text{firm}},\, h^{\text{firm}}\big).$$

The peak of ΔV_w is at the worker's bliss $\bar h = h^*$, with magnitude

$$\Delta V_w(h^*) = V_w(h^*) - V_w(h^{\text{firm}}) = \tfrac{\alpha}{2}(h^{\text{firm}} - h^*)^2 = \tfrac{2}{2}(0.20)^2 = 0.04$$

at calibration. Outside the band ($\bar h < 2h^* - h^{\text{firm}}$), the cap is so tight that the worker is forced past her bliss far enough to be worse off than at h^{firm}.

The two thresholds reconciled. The joint-surplus and worker-surplus bands have different lower edges:

$$\underbrace{2h^{\text{soc}} - h^{\text{firm}}}_{\text{joint turns negative}} \ge \underbrace{2h^* - h^{\text{firm}}}_{\text{worker turns negative}} \qquad (\text{since } h^{\text{soc}} \ge h^*).$$

Equality holds at the no-tax limit $\tau = 0$ (where $h^{\text{soc}} = h^*$ in this calibration); generically the joint threshold is strictly above the worker threshold. At baseline $(2h^{\text{soc}} - h^{\text{firm}},\, 2h^* - h^{\text{firm}}) = (0.30,\, 0.10)$, so for $\bar h \in (0.10, 0.30)$ the cap is welfare-reducing on aggregate (firm losses dominate) but the worker individually still benefits.

2 Formal derivation of the Q5 RTM bargain

2.1 Composite payoffs along the labour-demand curve

Per sector $j \in \{\text{G}, \text{B}\}$, the firm responds on its labour-demand curve $h_j^d(w) = (A_j - w)/\beta$. Substituting into the primitives:

$$\begin{aligned} V_w(w; A_j) &\equiv V_w(w, h_j^d(w)) = \frac{(1-\tau)w(A_j - w)}{\beta} - \frac{\alpha(A_j - w)^2}{2\beta^2}, \\ V_f(w; A_j) &\equiv V_f(w, h_j^d(w); A_j) = \frac{(A_j - w)^2}{2\beta}, \end{aligned}$$

where V_f collapses to the standard Cournot-residual form $(A_j - w)^2/(2\beta)$ because the firm is on its own FOC. Reparameterise $x_j \equiv A_j - w = \beta h_j^d(w)$; then

$$V_w(x_j) = A_v\, x_j - B_v\, x_j^2, \qquad A_v \equiv \frac{(1-\tau)A_j}{\beta}, \quad B_v \equiv \frac{1-\tau}{\beta} + \frac{\alpha}{2\beta^2}; \qquad V_f(x_j) = \frac{x_j^2}{2\beta}. \tag{5}$$

2.2 Lagrangian for the generalised Nash programme

Union and firm choose w to maximise the asymmetric Nash product subject to the participation (IR) constraints:

$$\max_w \left[V_w(w; A_j) - V_w^0\right]^\eta V_f(w; A_j)^{1-\eta} \qquad \text{s.t.} \quad V_w \ge V_w^0,\ V_f \ge 0.$$

Taking logs (both factors strictly positive on the interior), the Lagrangian with multipliers $\mu_w, \mu_f \ge 0$ is

$$\mathcal{L}(x_j; \mu_w, \mu_f) = \eta \ln\!\big(V_w(x_j) - V_w^0\big) + (1-\eta) \ln V_f(x_j) + \mu_w\big(V_w(x_j) - V_w^0\big) + \mu_f V_f(x_j),$$

using $w = A_j - x_j$ and $dw/dx_j = -1$ (so the FOC in x_j is the negative of the FOC in w, leaving the zero-locus unchanged). On the interior, $\mu_w = \mu_f = 0$ and the FOC is

$$\frac{\eta\, V_w'(x_j)}{V_w(x_j) - V_w^0} + \frac{(1-\eta)\, V_f'(x_j)}{V_f(x_j)} = 0, \qquad V_w' = A_v - 2B_v x_j,\ V_f' = x_j/\beta. \tag{6}$$

2.3 Reduction to a quadratic in x_j

Multiply (6) through by $\big(V_w - V_w^0\big)V_f$, substitute $V_f = x_j^2/(2\beta)$, and cancel one factor of $x_j/(2\beta)$ (assuming $x_j \neq 0$, which excludes the no-employment corner $w = A_j$):

$$\eta\,(A_v - 2B_v x_j)\, x_j + 2(1-\eta)\,\big(A_v x_j - B_v x_j^2 - V_w^0\big) = 0.$$

Collecting terms:

$$2B_v\, x_j^2 - A_v(2-\eta)\, x_j + 2(1-\eta)\, V_w^0 = 0. \tag{7}$$

The equilibrium $x_j^\star(\eta)$ is the larger root (the smaller root is the IR-binding fixed point at the $\eta = 0$ corner, see below), and the bargained wage and hours follow as $w_j(\eta) = A_j - x_j^\star(\eta)$, $h_j(\eta) = x_j^\star(\eta)/\beta$. At calibration $(\alpha, \beta, \tau, V_w^0) = (2, 2, 0.4, 0.05)$, (7) reads $1.1\, x^2 - 0.3A_j(2-\eta)x + 0.1(1-\eta) = 0$. Two checks: at $(A, \eta) = (2, 0)$ we get $x^\star = 1$, recovering $w = 1$, $h = 0.5$; at $(A_g, \eta) = (2.2, 0.5)$ we get $x^\star \approx 0.846$, giving $w \approx 1.354$, $h \approx 0.423$.

2.4 Boundary conditions

(i) $\eta = 0$**.** Equation (7) becomes $2B_v x_j^2 - 2A_v x_j + 2V_w^0 = 0$, i.e. $V_w(x_j) = V_w^0$. The worker's IR binds and the firm picks the wage that strips all rent above the outside option; on the firm's labour-demand curve this delivers $h_j = h_j^{\text{firm}}$.

(ii) $\eta \to 1$**.** (7) becomes $2B_v x_j^2 - A_v x_j = 0$, with non-trivial root

$$x_j^\star(1) = \frac{A_v}{2B_v} = \frac{(1-\tau)A_j\,\beta}{2\beta(1-\tau) + \alpha} \implies h_j(1) = \frac{x_j^\star(1)}{\beta} = \frac{(1-\tau)A_j}{2\beta(1-\tau) + \alpha}.$$

SOC at the limit. Evaluating the bracket $4B_v x_j - A_v(2-\eta)$ at $(x, \eta) = (A_v/(2B_v), 1)$ gives $4B_v \cdot A_v/(2B_v) - A_v = 2A_v - A_v = A_v > 0$, so the second-order condition for the Nash maximum holds at the limit and $x_j^\star(1)$ is indeed the maximiser, not a corner.

2.5 Walrasian benchmark and the comparative statics

Maintained regularity. Two conditions on V_w^0 are imposed throughout. *(R1)* $A_v^2 > 4B_vV_w^0$ (strict discriminant), so the IR-binding equation $V_w(x_j) = V_w^0$ admits two distinct real roots. *(R2)* $x_j^{\mathrm{FM}}(A_j)^2 > 2\beta^2V_w^0/\alpha$ over the range of A_j considered, which delivers $\partial w_j^{\mathrm{FM}}/\partial A_j > 0$. At calibration: (R1) $A_v^2 = 0.36 > 4B_vV_w^0 = 4(0.55)(0.05) = 0.11$. (R2) $x_j^{\mathrm{FM}} = 1$ and $2\beta^2V_w^0/\alpha = 0.2$, so (R2) holds with slack; on the sectoral range $A_j \in [1.8, 2.2]$ used in Q5 it remains satisfied (smallest value $x_b^{\mathrm{FM}} = 0.878$ gives $x^2 = 0.771 > 0.2$).

The competitive (free-market) outcome is the $\eta = 0$ specialisation of (7): $V_w(x_j) = V_w^0$, with larger root

$$x_j^{\mathrm{FM}}(A_j) = \frac{A_v + \sqrt{A_v^2 - 4B_vV_w^0}}{2B_v}, \qquad h_j^{\mathrm{FM}} = \frac{x_j^{\mathrm{FM}}}{\beta}, \qquad w_j^{\mathrm{FM}} = A_j - x_j^{\mathrm{FM}}. \tag{8}$$

Differentiating $V_w(x_j; A_j) = V_w^0$ implicitly and using $\partial A_v/\partial A_j = (1-\tau)/\beta$:

$$\frac{\partial x_j^{\mathrm{FM}}}{\partial A_j} = \frac{(1-\tau)\,x_j^{\mathrm{FM}}}{\beta(2B_vx_j^{\mathrm{FM}} - A_v)} > 0,$$

positive because the larger root sits past the vertex $x = A_v/(2B_v)$ of the parabola V_w, where $V_w' = A_v - 2B_vx < 0$. So $\partial h_j^{\mathrm{FM}}/\partial A_j = \beta^{-1}\partial x_j^{\mathrm{FM}}/\partial A_j > 0$. For the wage, substituting $A_vx_j = B_vx_j^2 + V_w^0$ into the numerator above gives $\partial x_j^{\mathrm{FM}}/\partial A_j = (1-\tau)x_j^2/[\beta(B_vx_j^2 - V_w^0)] < 1$ whenever $x_j^2 > 2\beta^2V_w^0/\alpha$ (using $\beta B_v - (1-\tau) = \alpha/(2\beta)$); this is exactly condition (R2), so $\partial w_j^{\mathrm{FM}}/\partial A_j = 1 - \partial x_j^{\mathrm{FM}}/\partial A_j > 0$. Together,

$$\frac{\partial h_j^{\mathrm{FM}}}{\partial A_j} > 0, \qquad \frac{\partial w_j^{\mathrm{FM}}}{\partial A_j} > 0,$$

i.e. "hours follow productivity" under perfect labour mobility.

2.6 Comparative statics

Differentiating (7) implicitly in η:

$$\big[4B_vx_j - A_v(2-\eta)\big]\frac{dx_j}{d\eta} = -A_vx_j + 2V_w^0.$$

Bracket positivity (global on $[0,1)$). The larger root of (7) is $x_j^\star(\eta) = [A_v(2-\eta) + \sqrt{D(\eta)}]/(4B_v)$ with $D(\eta) \equiv A_v^2(2-\eta)^2 - 16B_v(1-\eta)V_w^0$. At $\eta = 0$, $D(0) = 4A_v^2 - 16B_vV_w^0 = 4(A_v^2 - 4B_vV_w^0) > 0$ by (R1); at $\eta \to 1$, $D(1) = A_v^2 > 0$. $D(\eta)$ is a quadratic in η with positive leading coefficient A_v^2, vertex at $\eta_v = 2(1 - 4B_vV_w^0/A_v^2)$, and minimum value $D_{\min} = 16B_vV_w^0(A_v^2 - 4B_vV_w^0)/A_v^2 > 0$ under (R1). At calibration $\eta_v = 1.39$ lies outside $[0,1]$, so D is monotone on the relevant interval (decreasing from $D(0) = 1.0$ to $D(1) = 0.36$). Therefore $4B_vx_j^\star(\eta) = A_v(2-\eta) + \sqrt{D(\eta)} > A_v(2-\eta)$ on $[0,1)$, so the bracket $4B_vx_j - A_v(2-\eta) = \sqrt{D(\eta)} > 0$ strictly. The bracket equals $-V_w'(x_j^\star)\cdot$(positive) at the larger root, where $V_w'(x_j^\star) < 0$ (root past the vertex), so positivity of the bracket is the second-order condition for the Nash maximum.

Sign of the right-hand side. $-A_vx_j + 2V_w^0 < 0$ iff $A_vx_j > 2V_w^0$. At the larger root we have $x_j^\star(\eta) \geq x_j^\star(1) = A_v/(2B_v)$, so $A_vx_j^\star \geq A_v^2/(2B_v) > 2V_w^0$ where the last inequality is exactly (R1). So the RHS is strictly negative on $[0,1]$, and combined with the strictly positive bracket:

$$\frac{dx_j}{d\eta} < 0 \implies \frac{dw_j}{d\eta} = -\frac{dx_j}{d\eta} > 0, \qquad \frac{dh_j}{d\eta} = \frac{1}{\beta}\frac{dx_j}{d\eta} < 0,$$

the union-pulls-wages-up / hours-down comparative statics.

2.7 The welfare-restorer claim

Joint surplus. W is concave in h with peak at $h_j^{\mathrm{soc}} \in (h_j^*, h_j^{\mathrm{firm}})$. Since $h_j(\eta)$ is continuous and strictly decreasing in η with $h_j(0) = h_j^{\mathrm{firm}}$, by the intermediate-value theorem there exists a unique $\hat\eta_j \in (0,1)$ with $h_j(\hat\eta_j) = h_j^{\mathrm{soc}}$. Combining with the quadratic identity (3),

$$W\big(h_j(\eta)\big) - W(h_j^{\mathrm{firm}}) = -\tfrac{\alpha+\beta}{2}\Big[(h_j^{\mathrm{soc}} - h_j(\eta))^2 - (h_j^{\mathrm{soc}} - h_j^{\mathrm{firm}})^2\Big],$$

which is increasing on $[0,\hat{\eta}_j]$ (the union pulls hours *toward* the social bliss) and decreasing on $[\hat{\eta}_j,1]$ (overshoot region).

Worker surplus is monotone in η. On the labour-demand curve $V_w(x_j) = A_v x_j - B_v x_j^2$ is a concave parabola with maximum at $\hat{x} \equiv A_v/(2B_v) = x_j^\star(1)$. At $\eta = 0$, $V_w(x_j^\star(0)) = V_w^0$, so $x_j^\star(0)$ is the larger root of $V_w(x) = V_w^0$:

$$x_j^\star(0) = \frac{A_v + \sqrt{A_v^2 - 4B_v V_w^0}}{2B_v} > \frac{A_v}{2B_v} = \hat{x} = x_j^\star(1),$$

which together with monotonicity of $x_j^\star(\eta)$ gives $x_j^\star(\eta) \in (\hat{x}, x_j^\star(0)]$ for all $\eta \in [0,1)$. On this interval $V_w'(x) = A_v - 2B_v x < 0$, so as $x_j^\star$ falls in η, V_w rises:

$$\frac{dV_w}{d\eta} = V_w'(x_j^\star) \cdot \frac{dx_j^\star}{d\eta} > 0 \qquad \text{for all } \eta \in [0,1),$$

a product of two negatives.

Conclusion. Joint surplus is hump-shaped in η on $[0,1]$ with peak at $\hat{\eta}_j$, and worker surplus is strictly increasing in η. A planner with any positive worker weight $\lambda > 0$ in the Q7 weighting $W^\lambda(\eta) = (1+\lambda)V_w + (1-\lambda)V_f$ thus strictly prefers $\eta > 0$ to $\eta = 0$, since V_f is bounded above by its $\eta = 0$ value while V_w is strictly higher.